

Week 2 Design Studio

MAE 155A Group 2 - Ultra Fuel-Efficient 250 Passenger Aircraft

Sahil Savasere, Sainirnay Mantrala, Martina Danieli, Gregorio Zaltzman, Anthony Gaeta, Alexander Burns

Design Goal:

Design a 250 pax ultra fuel-efficient aircraft compatible with a code E gate

Design Choices:

- **Blended Wing Body**
- **Foldable Wing Tips**
- **Engine Type & Placement**

Design Requirements:

Mach	Range	Payload	Wingspan
~.78	~7,000 nm	~250 pax 50,000 Kg	<65m folded

Target Market and Assumptions

Target Market

- FSC Airlines with low average fleet age
 - Afford capital cost for diversification
 - Fuel efficiency
 - Premium comfort
- No budget Airlines due to diversification costs

Assumptions

- EIS: ~2040
 - Design: 10 years
 - Prototype & Manufacturing: 1-3 years
 - Testing & Certification: 1-2 years
- Compatible with current airport infrastructure & technology
- FAA certification achievable
 - Folding wings
 - Blended wing body

Design Choices

Blended wing body (BWB)

- Greater lifting surface (entire aircraft)
- Higher lift-to-drag ratio
- Decreased fuel burn per seat
- Decreased takeoff weight ([source](#))

Folding wingtip

- Longer in-flight wingspan while maintaining gate compatibility (Code E)
- Higher aerodynamic efficiency

Engine

- Ultra-High Bypass Turbofan
- Boundary Layer Ingestion

Design Requirements

Range: ~7000 nm

- Targeting FSC carriers flying these routes
- Increased fuel efficiency importance on higher fuel payloads

Cruise Speed: Mach 0.78-0.8

- Reference aircrafts (Mach 0.8-0.85)
- For BWB, Aerodynamic efficiency maximized at lower transonic mach numbers
- Lower speed reduces: wave drag, required cruise thrust, structural loads and aeroelastic penalties, specific fuel consumption

Folded Wingspan: <65m

- Fit within current airport infrastructure for competing similarly sized aircraft (Code E)

Competition & Reference Aircraft

Competing aircraft Pugh Chart
*Reference aircraft vs our Blended Wing
Body concept*

Criteria	A321neo	B787-8	JetZero Z4	Our BWB
Wingspan	0	+	+	++
Total Wing Area	0	+	++	++
Aerodynamic Efficiency	0	+	++	++
Fuel Burn/Seat	0	+	++	++
Payload-Range	-	+	+	+
Structural Complexity	0	0	--	---
Control Complexity	0	0	--	--
Maintenance Cost	++	+	-	--
Airport Compatibility	++	0	-	0
Passenger Comfort	0	+	++	++

Work Distribution

Name	Contributions
Sahil	Design Choice: Blended Wing Body, competition (Jetzero) research
Martina	Target Market, Design Requirements: Range
Sainirnay	Gantt Chart, Assumptions
Greg	Reference Aircraft, Pugh Chart, Terminal Restrictions, Folding wingtips
Anthony	Assumptions, Design Requirements: Speed
Alex	Gantt Chart, Task Tracker, Starting Initial Sizing, Project Manager

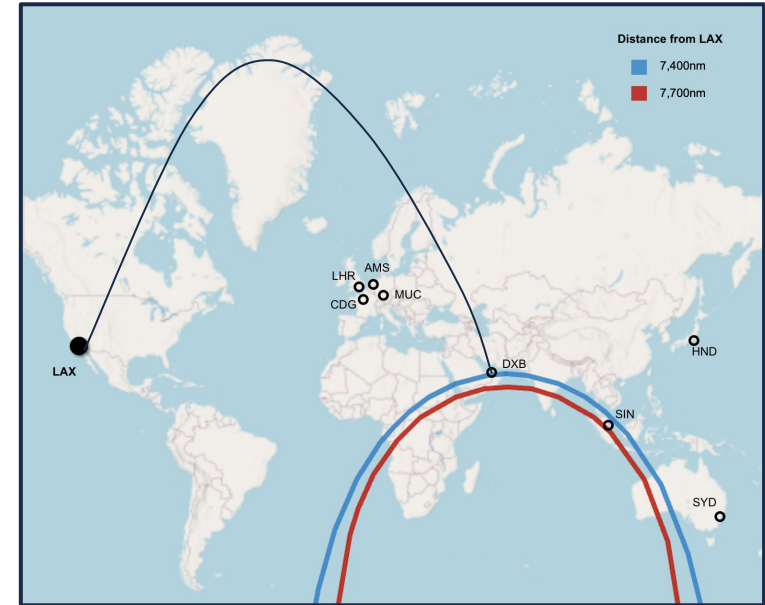
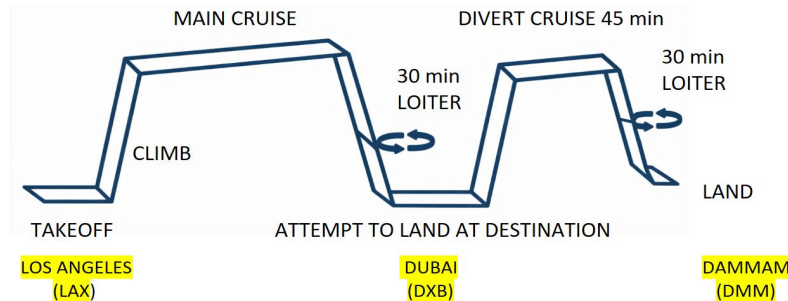
Week 3 Design Studio

MAE 155A Group 2 - Ultra Fuel-Efficient 250 Passenger Aircraft

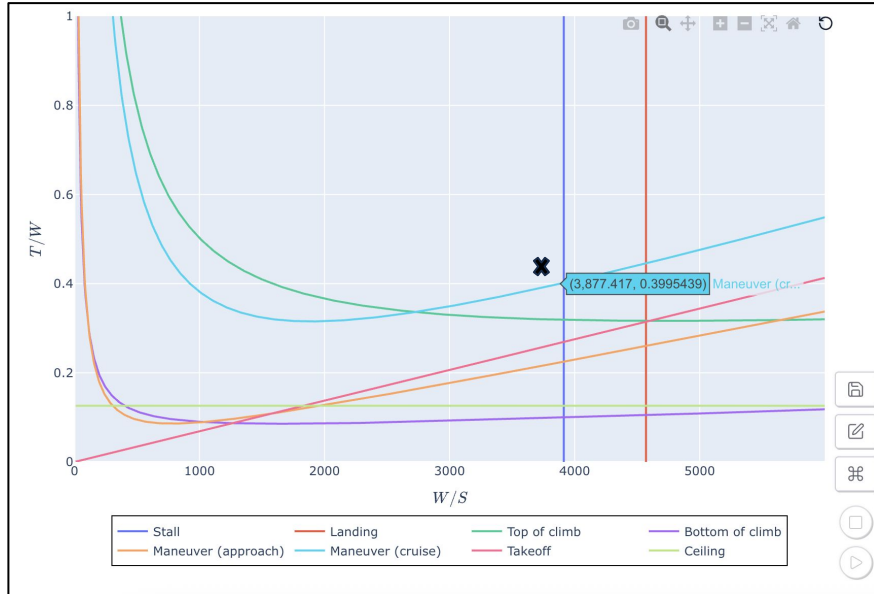
Sahil Savasere, Sainiray Mantrala, Martina Danieli, Gregorio Zaltzman, Anthony Gaeta, Alexander Burns

Mission profile

- Provide comfortable medium-long haul flights
- Speed: Mach 0.85
- Operating Route Range: ~7,400 nmi
- Operating Altitude: ~40,000 ft
- Example Route
 - LAX → DXB
- Fuel reserve complies with FAA/ICAO regulations (True cruise range of 10750 nmi)



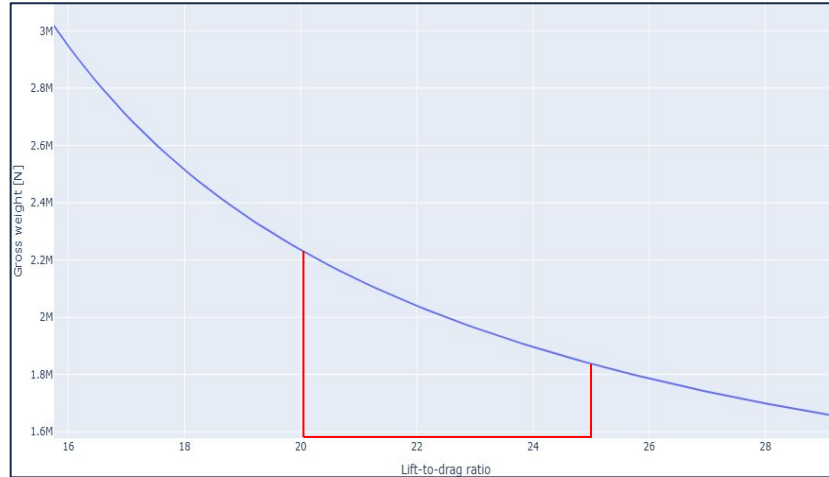
Initial Sizing



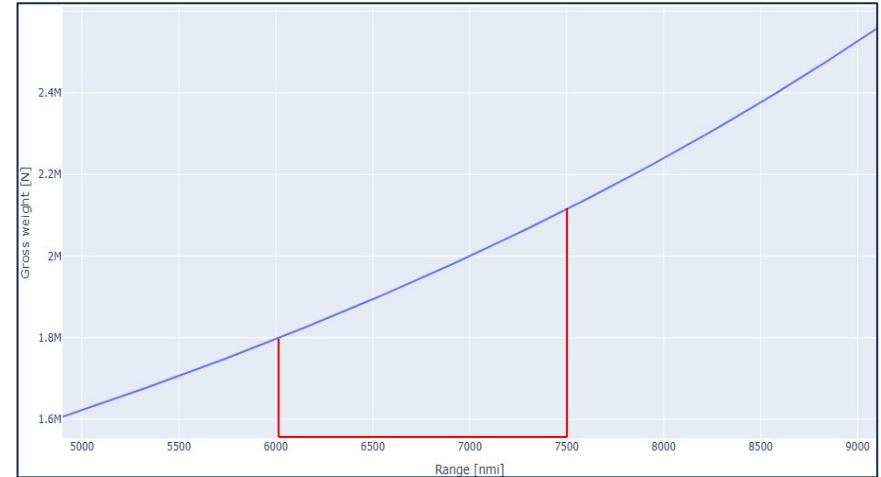
W/S and T/W include 5% margin from intersection

Gross weight [N]	1707321
Empty weight [N]	700080
Payload Weight [N]	392000
Energy storage weight [N]	615240
Empty weight fraction	0.41
Energy storage fraction	0.36
T/W	0.42
W/S [N/m^2 , kg/m^2]	3683, 375
Reference Area [m^2]	463.56
Thrust Required [kN]	717

Sensitivity Analysis



Parameter sweep: Gross weight vs L/D



Parameter sweep: Gross weight vs Range

Configurations

Embedded vs Non-embedded Engine

Parameter	Non-embedded	Embedded
Safety	0	--
Propulsive Efficiency	0	+
Aerodynamic Efficiency	0	+
Maintenance	0	-
Airflow intake	0	-
Heat management	0	-
Noise management	0	-
Space	0	-

Two vs Three Engines

Parameter	2 Engines	3 Engines
Redundancy	0	+
Efficiency	0	-
Yaw control (Engine out)	0	+
Structural Stability	0	-
Fuel efficiency	0	-
Noise	0	+
Maintenance cost	0	-

Vertical Stabilizer Arrangements

Parameter	No Vertical Stabilizer	Inclined Twin Tail	Vertical Twin Tail	Wingtip Mounted (FBW)
Directional Stability	0	++	+	+
Lateral Stability	0	++	+	+
Flight dynamics	0	++	+	---
Drag	0	--	--	0
Weight	0	--	--	--
Design complexity	0	--	--	---
Complete Control Failure	0	++	++	0

Work Distribution

Name	Contributions
Sahil	Configuration: Vertical Stabilizer
Martina	Mission Profile
Sainirnay	Engine pugh chart, Initial Sizing (Weight + Marimo modification)
Greg	Wing loading & Thrust / Weight (initial sizing), both engine pugh charts
Anthony	Mission Profile
Alex	Initial Sizing (Weight Fractions & W/S input parameters), Vertical Stabilizer Configuration

Appendix A: Weight Fraction Parameters

Sizing parameters

Aircraft type

Empty weight fraction regression parameters:

Coefficient, a

Exponent, b

Payload weight, W_p [N]

Range, R [m]

Cruise speed, V [m/s]

Lift-to-drag ratio, L/D

TSFC, c_T [kg/N/s]

BSFC, c_P [kg*s/W]

Efficiency, shaft to thrust power, η_s

Efficiency, battery to thrust power, η_b

Battery specific energy, e_b [J/kg]

Battery specific energy, e_b [J/kg]

Loiter Time, t_l [min]

Additional Cruise Time, t_c [min]

Appendix B: T/W & W/S Parameters

Sizing parameters

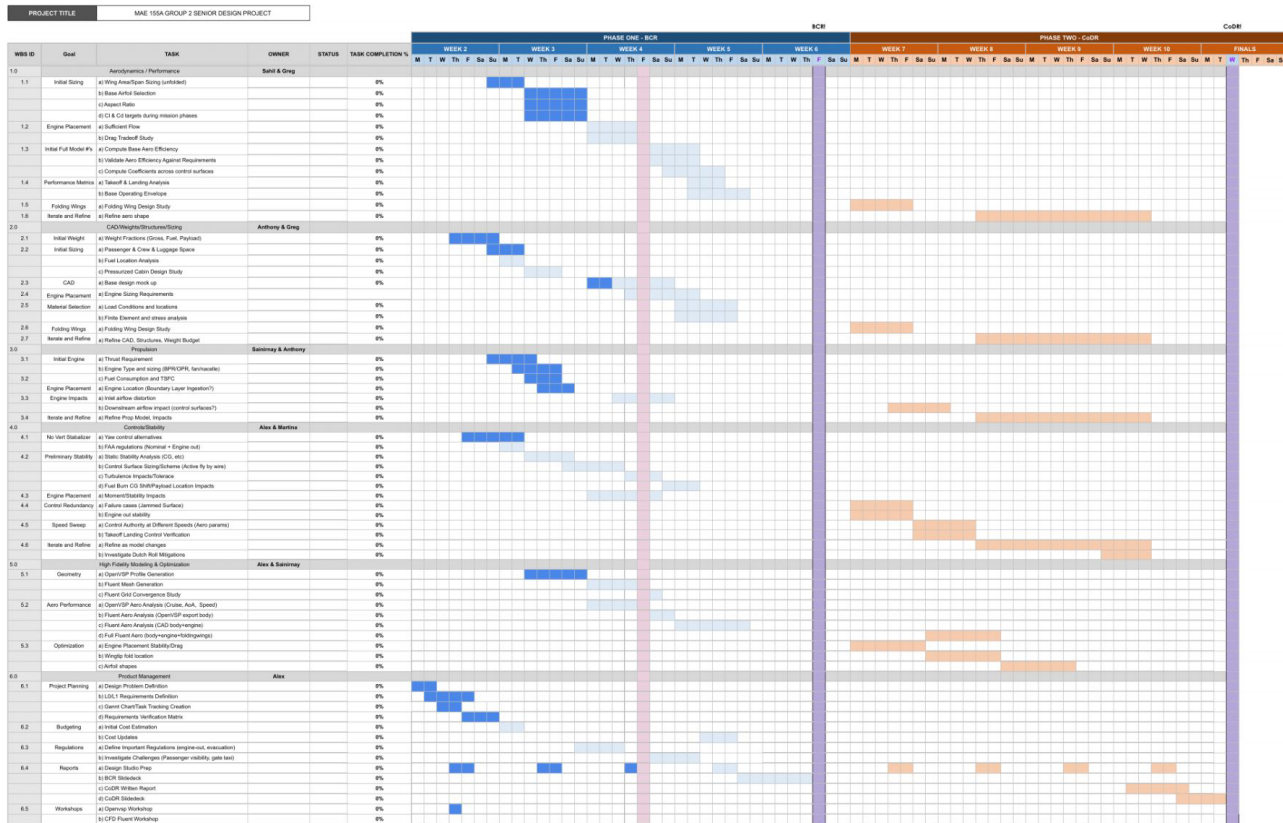
Cruise speed, V_{cruise} [m/s]	250.7	Climb speed, V_{climb} [m/s]	73.56	Stall speed, V_{stall} [m/s]	61.3
Takeoff speed, V_{takeoff} [m/s]	67.43	Approach speed, V_{approach} [m/s]	79.7		
Cruise altitude, h_{cruise} [m]	12192	Parasitic drag coeff., C_{D0}	0.0078		
Max. lift coeff., $C_{L_{\text{max}}}$	1.7	Takeoff lift coeff., $C_{L_{\text{takeoff}}}$	1.405		
Maneuver load factor, n	2.5	Climb angle, γ [deg]	2.7		
Oswald efficiency factor, e	0.85	Aspect ratio, AR	12		
Thrust Mach lapse, k ($T/T_{sl} = 1 - k \cdot M$)	0.4	Thrust density lapse, m ($T/T_{sl} = (\rho/\rho_{sl})^m$)	0.7		
Takeoff density altitude, h_{takeoff} [m]	1609.	Landing density altitude, h_{landing} [m]	1609.		
Takeoff parameter coefficient, k_{takeoff} [m ³ /N]	0.255	Landing parameter coefficient, k_{landing} [m ³ /N]	0.51		
Landing dist., d_{landing} [m]	1905	Airborne dist., d_{airborne} [m]	300.	Takeoff dist., d_{takeoff} [m]	3352.8

Week 4 Design Studio

MAE 155A Group 2 - Ultra Fuel-Efficient 250 Passenger Aircraft

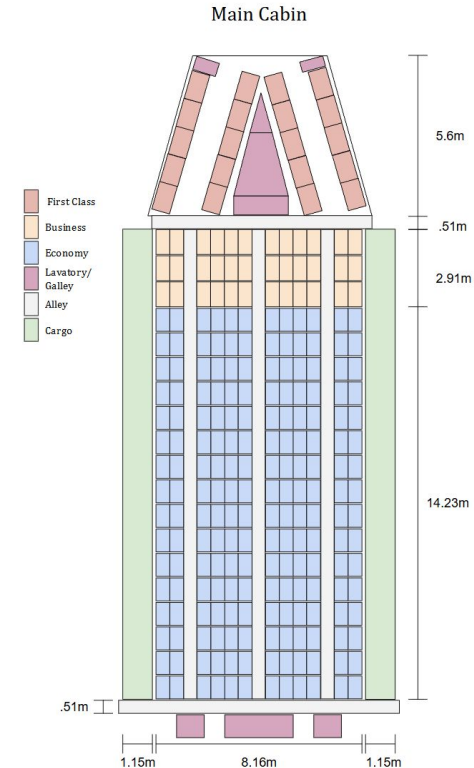
Sahil Savasere, Sainirnay Mantrala, Martina Danieli, Gregorio Zaltzman, Anthony Gaeta, Alexander Burns

Gantt Chart



Dimensions/ Configuration

Interior Cabin Width (m)	10.46
Exterior Cabin Width (m)	10.7132
Interior Cabin Length (m)	32.86
Total Central Chord Length (m)	48
Interior Cabin Height (m)	2.3
Total Fuselage Height (m)	5.76
Rows (bus, econ, tot)	3, 16, 19

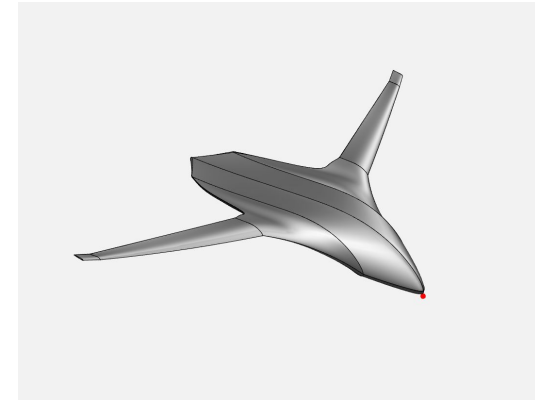


Main cabin configuration: 2-4-4-2

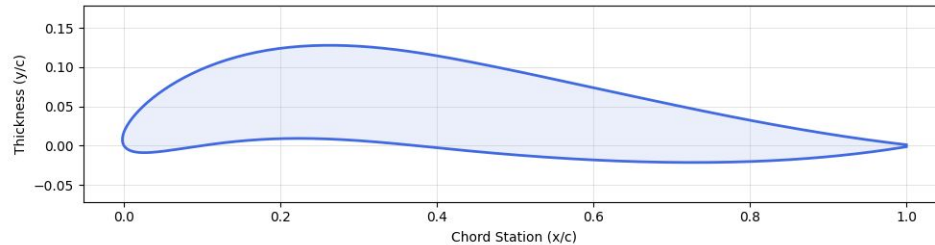
Airfoil Selection

OpenVSP model

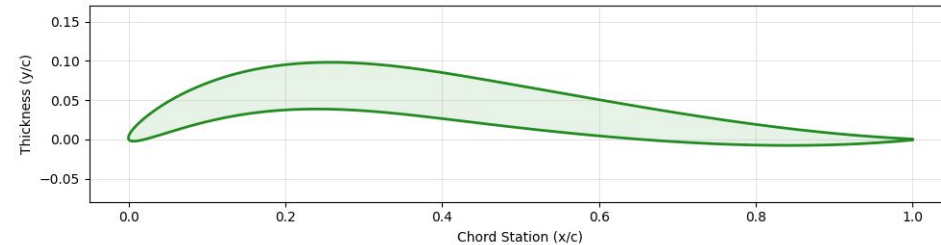
Section	Airfoil	$\frac{1}{2}$ Span (m)	Optimal Mach	t/c (%)
Inner	NACA 55112	10.58	$0.8 < M < .9$	12
Outer	NACA 55106	21.0	$0.8 < M < .9$	6



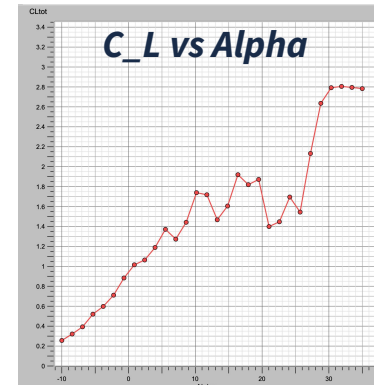
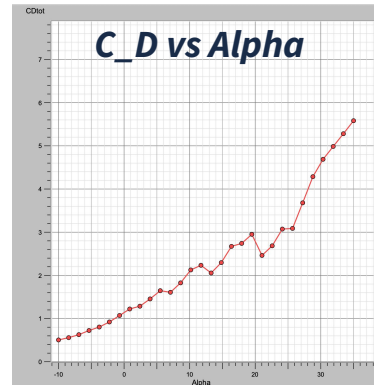
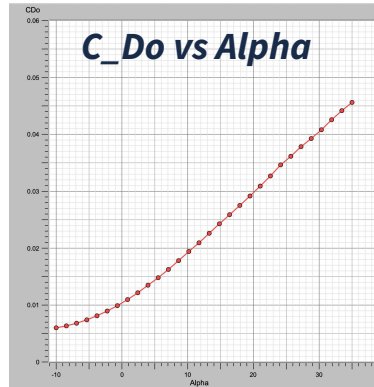
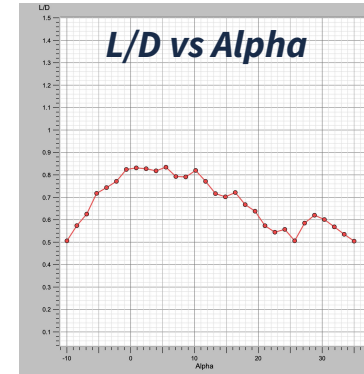
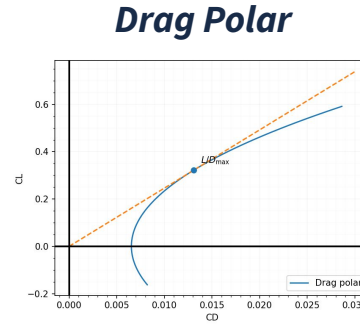
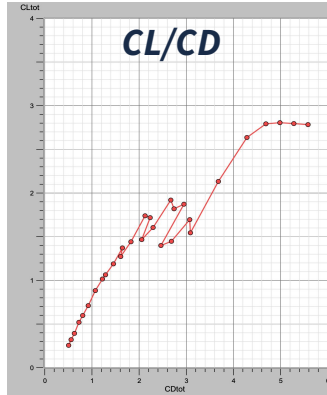
Inner Airfoil - naca55112



Outer Airfoil - naca55106



Performance Parameter Graphs



Performance Parameters

Empirical Calculations (Hwang Aerodynamics)

CL0	CL_max	CL_alpha	Parasitic Drag
0.2145	0.5915	0.0503	0.00653
Section	Airfoil	Critical Mach Number	Wave Drag Coef
Inner	NACA 55112	0.6685	0.0217
Outer	NACA 55106	0.7316	0.0039

OpenVSP Aero analysis

CL0	CL_max	CL_alpha	Parasitic Drag	Wave Drag Coef
0.94	1.9	0.07	0.00488	0.0370

Work Distribution

Name	Contributions,
Sahil	Performance Parameters
Martina	Airfoil Selection
Sainirnay	Performance Parameters
Greg	Dimensions/Configuration, Airfoil Selection, Performance Parameters
Anthony	Wave drag, Performance Parameters
Alex	Dimensions/Configuration, Airfoil Selection, Performance Parameters